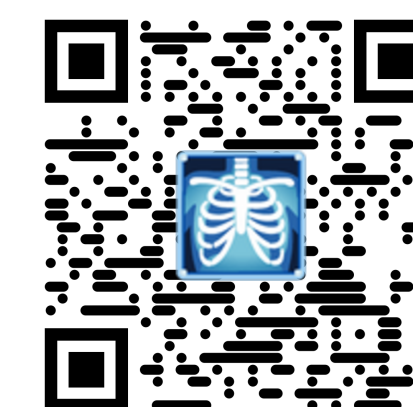




Overview

Radiology VLMs generate fluent reports, but they cannot point to the image regions behind their statements, so their output is hard to verify.

- Manual spatial annotation does not scale.
- Prior 2D CT slice datasets are small: Med-Gemini’s strict filtering yielded 4,009 images.

RadGrounder is a multi-task VLM for CT and MRI slices that performs report generation, VQA, and spatial grounding, trained without any manual spatial annotations.

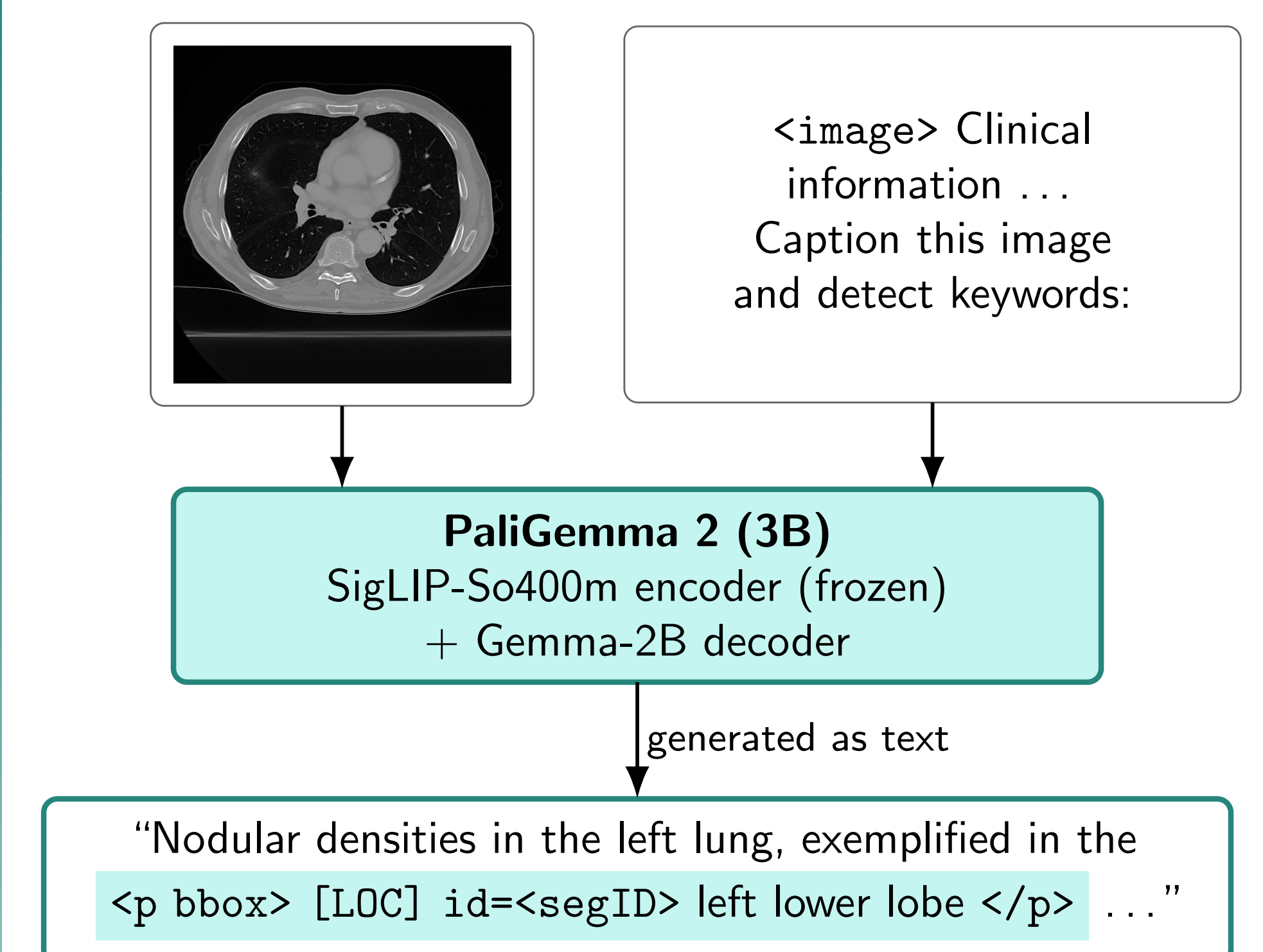
- Grounding supervision is derived automatically: TotalSegmentator masks linked to report text by LLM keyword extraction.
- Grounding does not degrade VQA or report quality.

RefRad2D dataset

RefRad2D: 1.27M image–text pairs from a decade of clinical routine at Uniklinik Freiburg.

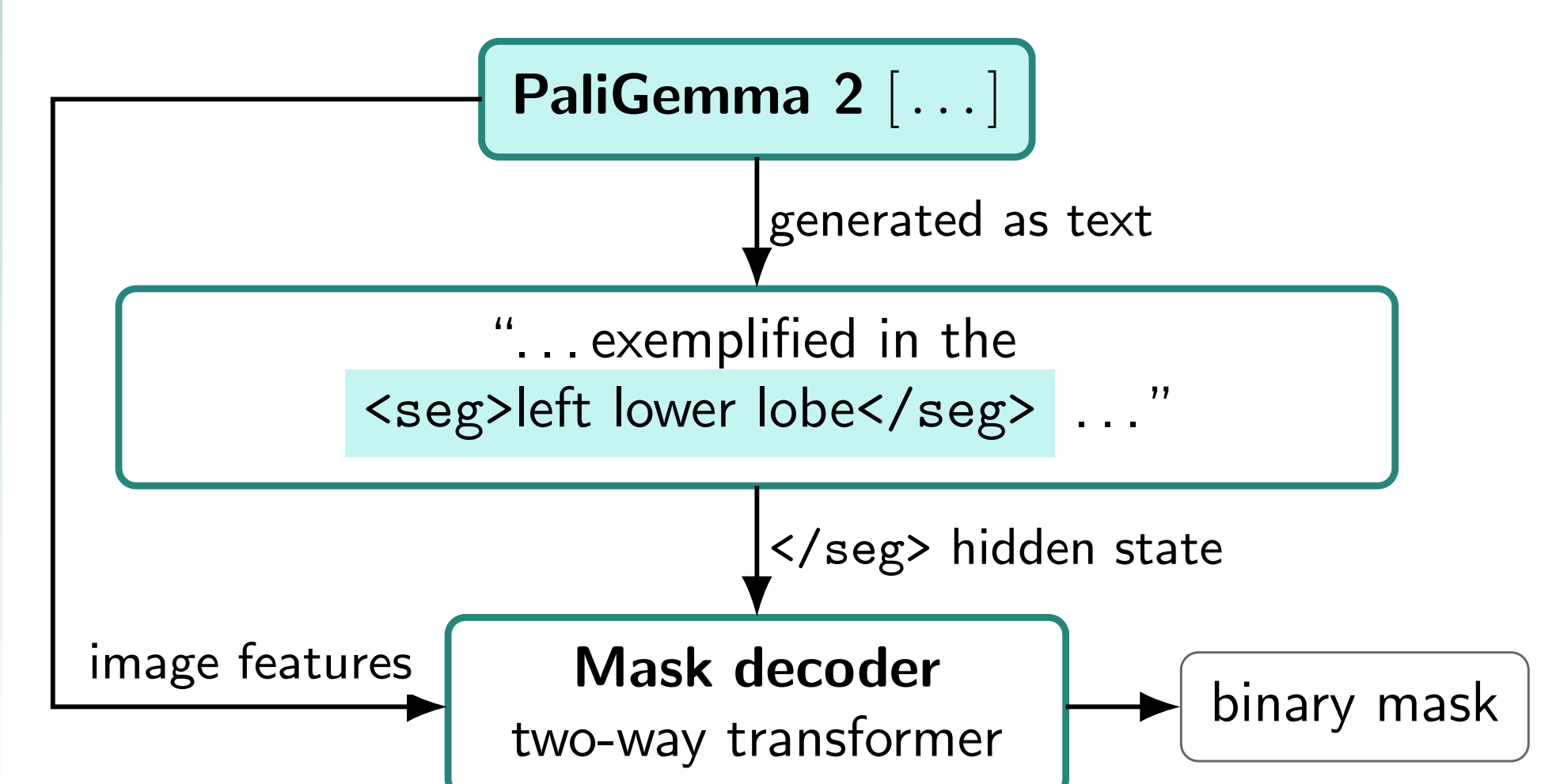
- We extracted report phrases with slice references such as “...renal cysts bilaterally, the largest on the right measuring 103×94 mm (Series 3, Image 50)”.
- 945k CT and 321k MRI slices
- LLM processing removes longitudinal references from captions and translates German to English.
- RefRad2D-Grounded: 236k slice–text–mask triplets. An LLM maps the anatomy named in the caption to the 121 TotalSegmentator classes. We keep a slice only if at least one of these structures is segmented in it.
- RefRad2D-VQA: ~9.6M synthetic QA pairs, 8 per image. Gemma 3 (27B) writes 5 from the findings (open, yes/no, multiple choice) and 3 from the slice metadata.

RadGrounder model



One model for report generation, VQA, and grounding. Detection is trained with next-token cross-entropy alone, since boxes are generated as tokens. The encoder is contrastively adapted to RefRad2D, then frozen.

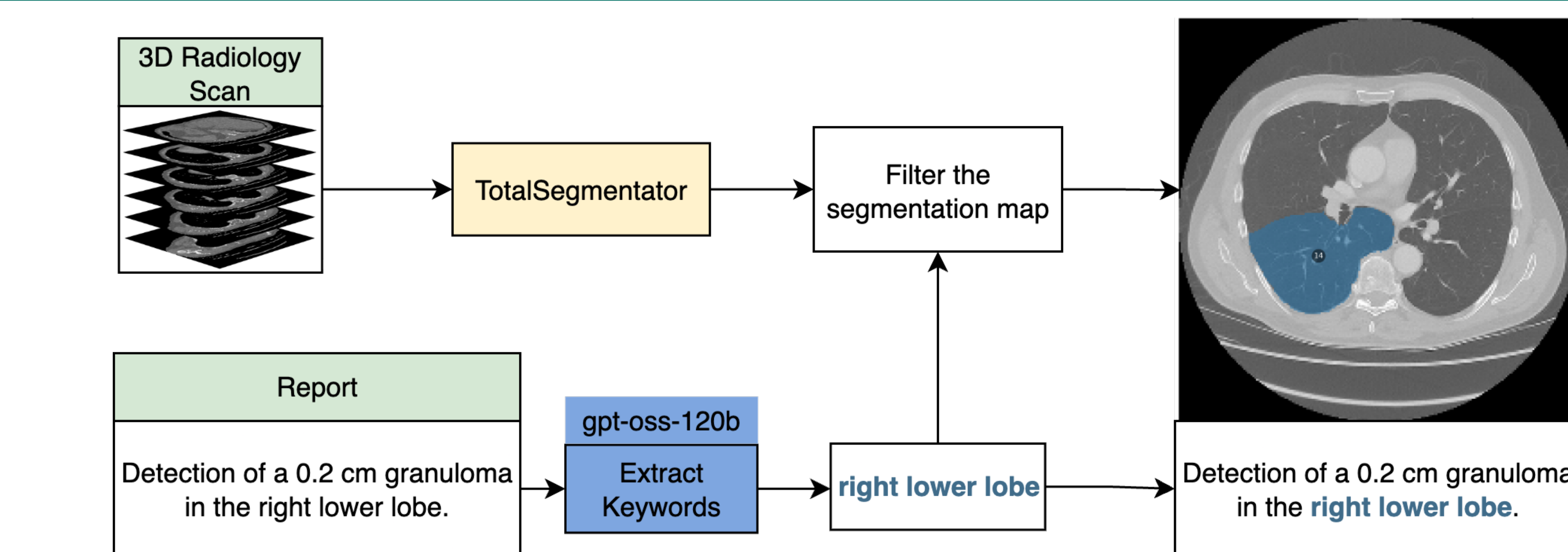
Segmentation variant: The hidden state of each generated `</seg>` token prompts a SAM-style mask decoder over the image features. Unlike detection, this adds an auxiliary decoder and focal + dice losses.



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References: Please refer to our paper for the complete list of references.

Automated spatial grounding



TotalSegmentator produces 3D segmentation masks, GPT-OSS (120B) extracts anatomical keywords from the report, and the segmentation map is filtered to the mentioned classes, yielding dense 2D annotations linked to the text.

Results

Grounding: Box detection localizes anatomical mentions without any extra decoder and achieves higher G-IoU than the segmentation head. Neither grounding strategy degrades VQA or report generation. → Spatially verifiable outputs at no cost to language quality.

Dataset transfer: Adding our corpus to external fine-tuning improves open-ended VQA on Slake and VQA-RAD on most metrics, and the resulting model is competitive with specialized medical VLMs. → Supervision from routine clinical reports transfers to external benchmarks.

Method	Report		VQA			G-IoU
	CIDEr	LLM	F1 _{op}	LLM _{op}	Acc _{cl}	
PaliGemma 2 (3B)	0.6	2.6	12.7	22.8	51.5	—
Gemma 3 (4B)	0.1	21.7	5.5	31.8	13.0	—
MedGemma (4B)	0.4	21.8	9.0	40.6	16.6	—
RadGrounder (Ours)	37.5	30.3	48.9	60.1	94.6	43.6
Segmentation head	36.9	30.1	50.4	64.1	95.6	36.9
No grounding	37.3	30.6	50.1	61.2	94.4	—
Internal data only	39.0	30.7	49.2	60.3	94.5	—
Internal + detect.	39.2	30.6	49.1	60.2	94.5	43.6
External VQA only	0.1	1.5	15.7	21.8	32.4	—

Internal RefRad2D test set: report captioning, VQA, and grounding (G-IoU). Subscripts: op = open-ended, cl = closed (yes/no and multiple choice). Bold marks values within the confidence interval of the best score, separately for baselines and ablations.

LLMScore: LLM-as-a-judge (Gemma 3, 27B) scores clinical factuality against the ground truth. On 200 VQA-RAD QA pairs scored by three radiologists, it correlates with their mean score at Pearson $r = 0.977$.

G-IoU: Matches predicted and ground-truth entities by text embedding similarity, then sums spatial IoUs weighted by that similarity, normalized to penalize both spatial hallucinations and omissions.

Method	Slake			VQA-RAD		
	F1	LLM _{op}	Acc _{cl}	F1	LLM _{op}	Acc _{cl}
PaliGemma 2 (3B)	24.5	34.2	58.1	37.9	26.0	55.6
Gemma 3 (4B)	40.2	22.3	53.0	33.6	11.1	33.6
MedGemma (4B)	72.3	54.9	87.6	49.9	20.8	69.1
BiomedGPT-B	85.2	—	89.9	*	—	*
LLaVA-Med	—	—	86.8	*	—	*
RadFM (14B)	84.4	—	—	*	—	—
Med-Gemini	75.8	—	84.8	*	—	78.8
RadGrounder (Ours)	87.7	88.8	90.3	50.7	44.3	64.7
Segmentation head	86.4	87.3	89.6	50.4	46.6	65.9
No grounding	87.2	88.8	90.8	49.2	42.8	63.2
External VQA only	86.9	87.8	88.4	47.9	28.6	68.1

External VQA benchmarks Slake and VQA-RAD, using the train/test split from Xu et al. due to train/test image overlap in the original VQA-RAD splits. *Reported results are not available for that split.

Qualitative examples

Report generation	Detection grounding	Segmentation grounding
Prompt: <code><image></code> Clinical information: Status post sleeve gastrectomy, bypass, with increasing abdominal pain. Question: Internal hernia? Caption this image: Ground Truth: Nonspecific increased mesenteric lymph nodes, up to 10mm. Prediction: Increased mesenteric lymph nodes up to 1.2 cm. Metrics: LLMScore: 75.0 CIDEr: 108.8	Prompt: <code><image></code> Clinical information: Severe pneumonia in a patient with COPD Gold 4. Question: Extent of the pneumonia? Caption this image and detect keywords: Ground Truth: Approximately 6 mm measuring round lesion in the left lower lobe. Prediction: Nodular densities in the left lung, exemplified in the left lower lobe with 5 mm, most likely postinflammatory. Metrics: LLMScore: 75.0 CIDEr: 13.7 G-IoU: 95.0	Prompt: <code><image></code> Clinical information: Breast carcinoma. Question: Malignant melanoma with unknown primary [...] Staging. Caption this image and segment keywords: Ground Truth: Small granuloma in the right upper lobe. Prediction: Small nodular opacity in the right upper lobe, most likely scar-like. Metrics: LLMScore: 50.0 mIoU: 61.0 G-IoU: 61.0